# FSP CAREER ARENA: PROJECT MANIFESTO

**Comprehensive Technical Documentation & System Specifications**

Reverse Recruiting, ELO Algorithms, 152-FZ Privacy, Judge0 Sandboxing

## proektnaya dokumentatsiya

### platforma obratnogo nayma s verifikatsiey FSP

> Komanda: NO PHP  NO PROBLEMS  2026
> Khakaton: Federatsiya sportivnogo programmirovaniya Rossii (FSP RF)  2026
> Versiya dokumenta: 1.0 | 10 oktyabrya 2026 g.

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## 1. Arkhitektura resheniya

### 1.1. Obshchee opisanie

Platforma FSP Karernaya Arena  eto tsifrovaya ekosistema iz tryokh klientskikh prilozheniy (Web, Mobile

### 1.2. Komponentnaya skhema



```
                              GlobalExceptionHandler
       Mobile App
      Kotlin 2.0              /api/v1/*
      Jetpack Compose         (Retrofit 2)        Integratsii:
      Material 3
      MVVM + StateFlow                              GitHub API (OAuth2)
                              HeadHunter API (OAuth2)
                                                    Yandeks ID (OAuth2)
                              LeetCode GraphQL Proxy
       Swagger UI             Judge0 Sandbox Engine
      OpenAPI v3 Spec         /swagger-ui.html


                                                   JDBC / JPA


                                                    PostgreSQL 16
                                                   Flyway: V1..V9
                                                   fsp_career_db

                                                   Tablitsy:
                                                    users
                                                    candidates
                                                    employers
                                                    vacancies
                                                    offers
                                                    algorithm_tasks
                                                    task_submissions
                                                    security_audit_logs
                                                    api_idempotency_records


                             LLM / AI Provaydery (II-portret)

       Gemini       OpenRouter      YandexGPT     Ollama (lokalnaya)
       Flash 1.5    Multi-model     Pro           Self-hosted LLM


                         Izolirovannaya Docker-pesochnitsa (Sandbox)
      Judge0 CE    Docker API    cgroups v2    seccomp    --net=none
```

### 1.3. Potoki dannykh

```
Soiskatel                  Platforma                    Rabotodatel

  1. Registratsiya /
     OAuth (GitHub/hh/Yandeks)
```

```
                                2. Verifikatsiya FSP ID
                                   + raschyot ELO

   3. Prokhozhdenie Areny

                                4. ELO pereschyot
                                   + II-portret

                                5. Poisk kandidatov
                                   po ELO/greydu/steku

                                6. Offer s vilkoy
                                   (kontakty skryty)

   7. Prinyatie / Otklonenie

                                8. Demaskirovka kontaktov
                                   (152-FZ audit log)
```

### 1.4. Stek tekhnologiy

| Sloy | Tekhnologii |
|---|---|
| Backend | Java 21 LTS, Kotlin 2.0, Spring Boot 3.3.4, Spring Security 6, Spring Data JPA, Hib |
| Frontend | React 19, TypeScript 7, Tailwind CSS 4, Vite 8, Monaco Editor, Lucide Icons, Canva |
| Mobile | Kotlin 2.0, Jetpack Compose, Material 3, Navigation Compose, Retrofit 2, Moshi, OkHt |
| BD | PostgreSQL 16 Alpine |
| Infra | Docker, Docker Compose, Nginx (SPA reverse proxy), GitHub Actions CI/CD, GHCR |
| AI/LLM | Google Gemini 1.5 Flash, OpenRouter (multi-model), YandexGPT Pro, Ollama (self-hoste |
| Sandbox | Judge0 CE, Docker Container Engine, Linux cgroups v2, seccomp |
| SSO (budushchee) | Keycloak 24 |

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## 2. Mekhanika obratnogo nayma i ranzhirovaniya

### 2.1. Kontseptsiya obratnogo nayma (Reverse Recruiting)

V klassicheskom nayme soiskatel otklikaetsya na vakansii. V FSP Karernoy Arene vsyo naoborot:

1. Rabotodatel izuchaet verifitsirovannyy profil sportsmena FSP (ELO-reyting, sportivnyy razryad, ste
2. Rabotodatel otpravlyaet offer s obyazatelnoy vilkoy zarabotnoy platy (v rublyakh, otdo).
3. Soiskatel prinimaet ili otklonyaet offer. Kontakty raskryvayutsya tolko posle prinyatiya.

> Klyuchevoy printsip: Rabotodateli konkuriruyut za talanty, a ne naoborot.

### 2.2. Algoritm matchinga

Ranzhirovanie kandidatov dlya rabotodatelya osushchestvlyaetsya po kompleksnoy formule CandidateMatchScore:

```
overallScore = w  fspRankScore + w  algoTestScore + w  stackMatchScore + w  experienceS
```

| Komponent | Ves | Opisanie |
|---|---|---|
| `fspRankScore` | 35% | Sportivnyy razryad FSP (ELO). Chem vyshe  tem bolshe ballov |
| `algoTestScore` | 25% | Rezultaty algoritmicheskogo testirovaniya v Arene (% prokhozhdeniya, vr |

| | | |
|---|---|---|
| `stackMatchScore` | 25% | Sovpadenie tekhnologicheskogo steka kandidata s trebovaniyami vakans |
| `experienceScore` | 15% | Kommercheskiy opyt raboty (gody, podtverzhdenie cherez hh.ru) |

### 2.3. Sistema ELO-reytinga i sportivnykh razryadov FSP

Reyting ELO privyazan k ofitsialnoy sisteme EVSK Minsporta RF po sportivnomu programmirovaniyu:

| Uroven | Razryad FSP | ELO diapazon | Pertsentil | Trebovaniya |
|---|---|---|---|---|
| 1 | Kvalifikatsionnyy divizion | 1200  1399 | Vne TOPa | Registratsiya v reestre FSP |
| 2 | 3-y sportivnyy razryad | 1400  1699 | TOP-35% | 1+ dostizhenie, 5+ zadach |
| 3 | 2-y sportivnyy razryad | 1700  1999 | TOP-15% | Prizovoe mesto, 25+ zadach |
| 4 | 1-y sportivnyy razryad | 2000  2399 | TOP-5% | 2+ dostizheniya, 100+ zadach |
| 5 | KMS (Kandidat v Mastera Sporta) | 2400  2699 | TOP-1.5% | Pobedy na turnirakh, 300+ |
| 6 | Master sporta RF | 2700  2999 | TOP-0.5% | 1-e mesto ChR, 500+ zadach |
| 7 | Grossmeyster / Chempion RF | 3000  3500 | TOP-0.05% | Mnogokratnyy chempion, 800+ za |

### 2.4. Raschyot ELO

Formula raschyota ELO iz ratingEngine.ts:

```
calculatedElo = 1200 (bazovyy start)
+ algoScore (Easy1 + Med2.5 + Hard6)
+ contestBonus ((contestRating - 1500)  0.35)
+ arenaBonus (solvedTasks  65)
```

Pri nalichii kontestnogo reytinga  1600:

```
targetElo = 0.35  targetElo + 0.65  contestRating
```

ELO ogranichen potolkom 3000 dlya areny i 2950 dlya importirovannykh rezultatov LeetCode.

### 2.5. Opredelenie greyda (Junior / Middle / Senior)

| Greyd | Usloviya |
|---|---|
| Senior | ELO  2400 ILI (opyt  5 let I ELO  1800) |
| Middle | ELO  2000 ILI (opyt  2 goda I ELO  1400) ILI opyt  4 goda |
| Junior | Ostalnye sluchai |

### 2.6. Antifrod-zashchita ranzhirovaniya

Kriticheski vazhnoe pravilo: kandidat s 0 dostizheniyami i 0 reshyonnymi zadachami ne mozhet popast v TOP n

- Bez istorii FSP  Kvalifikatsionnyy divizion (Bez istorii FSP)
- Est ELO, no 0 zadach i 0 turnirov  Vne TOPa (Kvalifikatsionnyy divizion)

### 2.7. Pravila skrytiya kontaktov (152-FZ)

| Etap | Vidimost kontaktov |
|---|---|
| Poisk kandidatov | Zamaskirovany: `+7 (9) *--42`, `@a*v` |
| Offer otpravlen | Zamaskirovany |
| Offer prinyat soiskatelem | Demaskiruyutsya  realnye dannye peredayutsya rabotodatelyu |
| Kazhdyy dostup | Zapis v `security_audit_logs` (kto, kogda, IP, tsel) |

### 2.8. Slepoy nayom (Anonymized / Blind Hiring Mode)

Dlya polnogo ustraneniya kognitivnykh iskazheniy i bessoznatelnoy predvzyatosti (Zero-Bias Standard) v p

- Maskirovanie identifikatorov: Imena kandidatov zamenyayutsya na standartizirovannyy psevdonim (nap
- Sokhranenie klyuchevykh obektivnykh metrik: Rekruter vidit isklyuchitelno obektivnye sportivnye dos
- Uslovie raskrytiya: Polnyy profil soiskatelya raskryvaetsya avtomaticheski posle vzaimnogo otklika

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# 3. Mekhanika testirovaniya i zashchity ot sliva

## 3.1. Algoritmicheskaya Arena FSP

Arena  vstroennaya sreda dlya resheniya algoritmicheskikh zadach s veb-IDE (Monaco Editor). Podderzhivaemye

Zadachi importiruyutsya iz tryokh istochnikov:
1. Vstroennyy bank zadach (klassicheskie LeetCode-zadachi, adaptirovannye pod FSP)
2. Zhivoy import iz LeetCode cherez GraphQL proksi
3. Kastomnye zadaniya ot rabotodatelya (cherez AssignTestModal)

## 3.2. Raschyot ELO za reshenie zadach

| Slozhnost | ELO bonus | Primer |
|---|---|---|
| Easy | +50 ELO | Two Sum, Valid Parentheses |
| Medium | +80 ELO | LRU Cache, Merge Intervals |
| Hard | +100 ELO | Median of Two Sorted Arrays |

## 3.3. Ogranichenie kuldauna peresdachi

- Polnaya peresdacha vsekh testov vozmozhna odin raz v 3 mesyatsa (pole `canRetakeAfterDate` v profile
- Otdelnye novye zadachi mozhno reshat bez ogranicheniy.
- Servernyy taymer blokiruet povtornuyu otpravku resheniya po uzhe sdannoy zadache.

## 3.4. Izolirovannaya Docker-pesochnitsa (Sandbox)

Kazhdaya posylka koda vypolnyaetsya v odnorazovom konteynere s polnoy izolyatsiey:

```
[ Monaco IDE ]  [ Spring Boot 3 Backend ]  [ Sandbox Manager ]

Docker API / cgroups v2 / seccomp


  IZOLIROVANNAYa DOCKER-PESOChNITsA
   --net=none (net seti)
   Memory: 128256 MB (cgroups)
   CPU: 10002000 ms na test
   User: nobody (non-root)
   FS: Read-Only rootfs
   PIDs Limit: 32
   seccomp: blocked syscalls


[ Servernyy Verdict Validator ]
(ACCEPTED / WA / TLE / CE / RE)
```

## 3.5. Parametry izolyatsii

| Parametr | Znachenie | Naznachenie |
|---|---|---|
| Limit sessii | 45 minut | Zashchita ot rastyagivaniya |
| CPU Time | 10002000 ms / test | Otsechka neoptimalnykh O(N) resheniy |
| RAM | 128256 MB | Zashchita ot OOM khosta |
| Set | `--net=none` | Isklyuchenie curl / LLM / vneshnikh BD |
| Faylovaya sistema | Read-Only + tmpfs 16MB | Zashchita ot vredonosnykh skriptov |
| Protsessy | PIDs  32 | Zashchita ot fork-bomb |
| Prava | `nobody:65534`, `CAP_DROP_ALL` | Isklyuchenie eskalatsii privilegiy |

### 3.6. Antifrod-kontrol

1. Skrytye test-keysy (Hidden Suites): Brauzer poluchaet tolko 2-3 otkrytykh primera. Granichnye i nag
2. Telemetriya chestnosti:
  - Otslezhivanie `window.blur` (poterya fokusa)
  - Inspektsiya `onPaste` (massivnaya vstavka chuzhogo koda)
  - Fiksatsiya anomaliy v `security_audit_logs`
3. Kriptograficheskaya fiksatsiya:
  - SHA-256 khesh iskhodnogo koda v tablitse `task_submissions`
  - Neizmenyaemye zapisi: vremya vypolneniya, potreblenie pamyati, taymstemp
4. Dinamicheskaya generatsiya zadach: Bank zadach rasshiryaetsya cherez live-import iz LeetCode API. Rabotodat

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## 4. Skhema integratsii s FSP ID

### 4.1. Tekushchaya realizatsiya

Na tekushchem etape profil kandidata soderzhit:
- `fspId` unikalnyy identifikator v reestre FSP RF (naprimer, `FSP-RUS-2026-0088`)
- `fspRank` sportivnyy razryad EVSK
- `fspRating` ELO-reyting
- `isVerified` status verifikatsii
- `keycloakSub` identifikator Keycloak (podgotovlen v skheme dannykh)
- `fspProfileUrl` ssylka na profil sportsmena na `fsp-russia.ru`

### 4.2. Arkhitekturnyy podkhod k budushchey integratsii s Keycloak

```
  Polzovatel                  Keycloak 24
  (Brauzer / Mobile)    Realm: fsp-arena
                      OIDC
                      flow      Klienty:
            web-frontend
   mobile-app
   backend-api

  Roli:
   CANDIDATE
   EMPLOYER
   ADMIN

  Federatsiya identiti:
   fsp-russia.ru IDP
   GitHub OAuth2
   Yandeks ID
   hh.ru OAuth2


JWT (RS256)



  Spring Boot 3
  Backend
```

```
   Spring Security 6
   OAuth2 Resource
   Server

    Validatsiya JWT
    Mapping roley
    Sinkhronizatsiya
     fspId  keycloakSub
```

Shagi migratsii:

1. Razvernut Keycloak 24 kak otdelnyy servis v docker-compose.yml
2. Sozdat Realm fsp-arena s rolyami CANDIDATE, EMPLOYER, ADMIN
3. Nastroit Identity Federation s reestrom FSP (fsp-russia.ru) cherez SAML/OIDC
4. Pereklyuchit Spring Security s sobstvennogo JWT na OAuth2 Resource Server s validatsiey JWT ot Keyc
5. Mappirovat keycloakSub na fspId pri pervoy autentifikatsii

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## 5. Spetsifikatsiya API i bezopasnost

### 5.1. Swagger UI i OpenAPI

| Resurs | URL |
|---|---|
| Swagger UI (interaktivnaya dokumentatsiya) | `http://localhost:8080/swagger-ui.html` |
| OpenAPI v3 JSON (mashinnaya spetsifikatsiya) | `http://localhost:8080/v3/api-docs` |
| Oblachnyy stend | `https://linskay-demo-2026.hf.space` |

### 5.2. Osnovnye endpointy API (`/api/v1/`)

| Metod | Endpoint | Opisanie | Rol |
|---|---|---|---|
| `POST` | `/auth/login` | Autentifikatsiya (email + parol  JWT) | Vse |
| `POST` | `/auth/register` | Registratsiya novogo polzovatelya | Vse |
| `GET` | `/auth/test-users` | Spisok testovykh akkauntov dlya 1-klik vkhoda | Vse |
| `GET` | `/auth/privacy-policy` | Politika konfidentsialnosti (152-FZ) | Vse |
| `GET` | `/users/me` | Dannye tekushchey sessii polzovatelya | Vse |
| `PUT` | `/users/profile` | Obnovlenie profilya polzovatelya | Vse |
| `GET` | `/candidates` | Unifitsirovannyy poisk i filtratsiya kandidatov | EMPLOYER |
| `GET` | `/candidates/{id}/pdf` | Avtogeneratsiya ofitsialnogo PDF-pasporta FSP | Vse |
| `GET` | `/vacancies/search` | Poisk vakansiy s obyazatelnoy vilkoy zarplat | CANDIDATE |
| `POST` | `/vacancies` | Sozdanie vakansii | EMPLOYER |
| `POST` | `/sandbox/execute` | Vypolnenie koda v pesochnitse (Judge0) | CANDIDATE |
| `POST` | `/leetcode/fetch/{slug}` | Import zadachi iz LeetCode po slug | Vse |
| `POST` | `/ai/generate-summary` | Multimodelnaya generatsiya II-portreta (Strategy) | EMPL |
| `POST` | `/ai/save-profile` | Sokhranenie II-rezyume v profil kandidata | EMPLOYER |
| `POST` | `/offers` | Otpravka offera (s Idempotency-Key i Webhook) | EMPLOYER |
| `POST` | `/webhooks` | Registratsiya iskhodyashchego veb-khuka dlya ATS | EMPLOYER |
| `GET` | `/webhooks/employer/{id}` | Spisok aktivnykh veb-khukov rabotodatelya | EMPLOYER |
| `POST` | `/webhooks/{id}/ping` | Testovaya otpravka pinga dlya proverki veb-khuka | EMPLOYER |
| `POST` | `/files/upload` | Multipart zagruzka rezyume i dokumentov | Vse |
| `GET` | `/system/stats` | Otkrytaya statistika platformy | Vse |
| `POST` | `/auth/sync/github-profile` | Sinkhronizatsiya resheniy iz GitHub | CANDIDATE |
| `POST` | `/auth/sync/hh-resume` | Sinkhronizatsiya rezyume hh.ru | CANDIDATE |
| `GET` | `/audit/logs` | Zhurnal audita bezopasnosti (152-FZ) | EMPLOYER |

### 5.3. Rolevaya model (JWT)

```
Authorization: Bearer <JWT>
```

| Rol | Opisanie | Razresheniya |
|---|---|---|
| `CANDIDATE` | Soiskatel / Sportsmen FSP | Prokhozhdenie Areny, prosmotr offerov, upravlenie |
| `EMPLOYER` | IT-kompaniya / HR | Poisk kandidatov, sozdanie vakansiy, otpravka offerov, nazn |

JWT soderzhit: userId, email, fullName, role, candidateId / employerId, fspElo, fspRank.

### 5.4. Idempotentnost API (RFC 7231)

Pri otpravke offera (POST /api/v1/offers):

1. Klient generiruet zagolovok Idempotency-Key: idem-<uuid>
2. Bekend proveryaet tablitsu api_idempotency_records:
   - Pervyy zapros: Sozdayot offer, sokhranyaet klyuch `HTTP 201 Created`
   - Povtornyy zapros (retray): Vozvrashchaet sokhranyonnyy offer `HTTP 200 OK` + zagolovok `Idempotent
3. Unikalnyy indeks (employer_id, candidate_id, vacancy_id) WHERE status = 'PENDING' isklyuchaet dubl

### 5.5. Sootvetstvie 152-FZ

| Mera | Realizatsiya |
|---|---|
| Maskirovanie PDN | Telefon, email, Telegram zamaskirovany v DTO (`+7 (9) *--42`) |
| Soglasie polzovatelya | Obyazatelnyy chekboks pri registratsii, ssylka na [Polozhenie FSP o PDN] |
| Demaskirovka po soglasiyu | Kontakty raskryvayutsya tolko posle prinyatiya offera kandidatom |
| Zhurnal audita | Tablitsa `security_audit_logs`: actorUserId, actorEmail, targetCandidateId, ip |
| Dostup k auditu | `GET /api/v1/audit/logs` + modalnoe okno `AuditLogModal` v UI |

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## 6. Instruktsiya po razvyortyvaniyu

### 6.1. Sistemnye trebovaniya

| Komponent | Minimum |
|---|---|
| Docker | 24+ |
| Docker Compose | v2+ |
| RAM | 4 GB |
| Disk | 5 GB |
| OS | Windows 10+, macOS 12+, Ubuntu 22.04+ |

### 6.2. Bystryy start (odna komanda)

Linux / macOS:

```
git clone https://github.com/linskay/fsp-2026.git
cd fsp-2026
chmod +x start.sh
./start.sh
```

Windows (cmd ili PowerShell):

```
git clone https://github.com/linskay/fsp-2026.git
cd fsp-2026
start.bat
```

Ili napryamuyu cherez Docker Compose:

```
docker compose up -d
```

### 6.3. Peremennye okruzheniya (.env)

Skopiruyte .env.example v .env i pri neobkhodimosti zamenite znacheniya:

```
cp .env.example .env
```

| Peremennaya | Opisanie | Po umolchaniyu |
|---|---|---|
| `GEMINI_API_KEY` | API-klyuch Google Gemini (II-portret) | Predustanovlen |
| `OPENROUTER_API_KEY` | API-klyuch OpenRouter (multi-model LLM) | Predustanovlen |
| `OLLAMA_API_URL` | URL lokalnogo Ollama servisa | `http://localhost:11434/...` |
| `YANDEX_API_KEY` | API-klyuch YandexGPT | (optsionalno) |
| `YANDEX_FOLDER_ID` | Folder ID Yandex Cloud | (optsionalno) |
| `APP_URL` | Adres khostinga | `http://localhost:3000` |
| `GITHUB_CLIENT_ID` | GitHub OAuth2 Client ID | mock-fallback |
| `GITHUB_CLIENT_SECRET` | GitHub OAuth2 Client Secret | mock-fallback |
| `HH_CLIENT_ID` | hh.ru OAuth2 Client ID | mock-fallback |
| `HH_CLIENT_SECRET` | hh.ru OAuth2 Client Secret | mock-fallback |
| `JUDGE0_API_URL` | URL Judge0 Sandbox API | Predustanovlen |

### 6.4. Docker Compose servisy

| Servis | Port | Opisanie |
|---|---|---|
| `postgres` | 5432 | PostgreSQL 16 Alpine |
| `backend` | 8080 | Spring Boot 3 (Java 21) |
| `frontend` | 3000 | React 19 + Nginx |
| `ollama` | 11434 | Lokalnaya LLM (optsionalno) |

### 6.5. Proverka rabotosposobnosti

Posle zapuska otkroyte:

1. Frontend: [http://localhost:3000](http://localhost:3000) glavnaya stranitsa
2. Swagger UI: [http://localhost:8080/swagger-ui.html](http://localhost:8080/swagger-ui.html) API
3. Healthcheck: curl http://localhost:8080/api/v1/auth/privacy-policy status bekenda

### 6.6. Testovye uchyotnye zapisi

| Rol | Email | Parol |
|---|---|---|
| Kandidat | `candidate@fsp.ru` | `candidate2026` |
| Rabotodatel | `employer@fsp.ru` | `employer2026` |

### 6.7. Ostanovka platformy

```
docker compose down          # Ostanovit konteynery
docker compose down -v       # + udalit dannye (volumes)
```

---

## 7. Komanda proekta

### NO PHP  NO PROBLEMS  2026

Proekt razrabotan v ramkakh ofitsialnogo khakatona Federatsii sportivnogo programmirovaniya Rossii.

| Zona otvetstvennosti | Opisanie |
|---|---|
| Arkhitektura & Backend | Spring Boot 3, migratsii PostgreSQL 16 Flyway, Kotlin 2.0 + Java 21 |
| Frontend & UX/UI | React 19 + Vite, Code Arena IDE, integratsiya s GitHub API, OAuth2 avtori |
| Mobile App (Android) | Kotlin 2.0 + Jetpack Compose + Material 3, Clean Architecture + MVV |
| DevOps & AI Engine | CI/CD na GitHub Actions, multiprovaydernyy II-skoring, deploy na Hug |

### komandy

1. Strogaya staticheskaya tipizatsiya  Java 21, Kotlin 2.0, TypeScript 7
2. Chistaya arkhitektura  polnoe razdelenie sloyov vo vsekh klientakh
3. Bezopasnost i zakon  152-FZ, maskirovanie PDN, audit
4. Inzhenernaya estetika  firmennyy stil FSP, JetBrains Mono

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> NO PHP  NO PROBLEMS  2026  sozdayom tsifrovoe budushchee sportivnogo programmirovaniya Rossii!